Investigation Report

A-Mab Drug Substance — INV-001

Quality Assurance (QA)

2026-09-18

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Approvals

Role	Function	Name (synthetic)	Signature	Date
Author	Quality Assurance (investigator)	—	—	—
Reviewer	Manufacturing Science & Technology	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Statistics	—	—	—
Approver	Head of Quality	—	—	—

Synthetic record. Signatures are not applied.

Revision history

Version	Effective date	Description of change
1.0	2026-09-18	Initial issue

Abbreviations

Abbreviation	Meaning
AEX	anion exchange chromatography
BLA	Biologics License Application
CAPA	corrective and preventive action
CEX	cation exchange chromatography
CI	confidence interval
CQA	critical quality attribute
DoE	design of experiments
DS	drug substance
GMP	good manufacturing practice
HCP	host cell protein
HMW	high molecular weight species
IAC	intermediate acceptance criterion
LFC	least favourable combination of the other parameters
LMW	low molecular weight species
LRF	log reduction factor
MSAT	Manufacturing Science & Technology
MVM	minute virus of mice
NOR	normal operating range
nrCE-SDS	non-reduced capillary electrophoresis with sodium dodecyl sulfate
PAR	proven acceptable range
QA	Quality Assurance
SD	standard deviation
SEC	size-exclusion chromatography
SOP	standard operating procedure
TFF	tangential flow filtration
UF/DF	ultrafiltration and diafiltration
VI	viral inactivation
XMuLV	xenotropic murine leukemia virus

Summary

On 2026-08-03, Quality Assurance opened this investigation, INV-001, during the review of the Process Characterization Master Report PCMR-001 Version 1.0 for BLA readiness. The review found that the acceptance criteria against which each step was judged in the proven acceptable range analysis were not in any protocol approved before the studies ran. They had been set at analysis time, after the data were known. The investigation was classified as major.

The investigation makes six findings, numbered F1 to F6 in §4. Four concern the criteria themselves. The criteria were defined after the data were seen (F1). For impurities that downstream steps clear, each criterion was divided by a safety factor that differed by step, and the factors were such that every step passed at its own setpoint (F2). The glycan and acidic variant limits at the bioreactor were computed from the capability data they were then used to judge (F3). Acidic variants were assessed against the upper end of their two-sided range only (F4). Finding F5 concerns the records. Their title blocks misstated the dates on which the plans were approved, and they gave no execution dates. Finding F6 concerns the method for fragments. SEC read fragments high in the Protein A pool and the low-pH viral inactivation pool, and the decision AMD-001 to measure fragments by nrCE-SDS instead was not carried into the documents.

The root cause is that the Process Characterization Master Plan PCMP-001 Version 1.0 required a criterion for each step but gave no rule for deriving one. The criteria were therefore derived during the analysis, by the people analysing the data (§5).

Under corrective action CA1, the investigation defined intermediate acceptance criteria by a rule written before any report applied it. Each criterion is carried back from the drug-substance criterion through the platform worst case of each downstream step, with no margin. This campaign’s data can move a criterion only through the one-sided 95 % prediction bound of a single batch of a downstream step, at the least favourable combination within its normal operating range (NOR). The same rule set fixes how a scale-down model whose mean differs from manufacturing scale is qualified, and that viral clearance is judged at the least favourable combination of the other parameters. Corrective action CA1 issues all of this in PCMP-001 Version 2.0. Corrective actions CA2 and CA3 reissue the step plans and every other document of the package against it, with fragments measured by nrCE-SDS. They are due on 2026-09-30 and were open at closure. Preventive action PA1 revises SOP-1002 so that a characterization protocol cannot be approved without its criteria (§7).

Of the 61 proven acceptable ranges that Version 1.0 reported with the other parameters varying within their NORs, 29 are wider in the re-analysis, 3 are narrower, 3 have moved, 3 no longer give a range and 23 are unchanged. In the re-analysis, acidic variants at the production bioreactor step and XMuLV clearance at the low-pH viral

inactivation step do not meet their Version 2.0 criterion across the whole NOR, although each meets it with every parameter at its set-point (§6.3). At Version 2.0, the Protein A report PCR-005 keeps the NOR in force and flags it for discussion with manufacturing, because over that NOR the scale-down model is not qualified for pool host cell protein (§2). The low-pH viral inactivation report PCR-006 moves the NOR and the set-point of inactivation pH lower and narrows the NORs of hold time and temperature, because XMuLV clearance does not meet its claim at the least favourable combination of the NOR as it was characterized (§6.3). The lowest capability index of the drug substance is now that of acidic variants, at 1.03, against its lower limit. No GMP batch was dispositioned on these documents, and there is no product impact (§6.6).

1 Event description

PCMR-001 Version 1.0, the Process Characterization Master Report, and the step reports it consolidates were approved on 2026-07-24. On 2026-08-03, QA reviewed PCMR-001 for use in the BLA. The review traced the in-process criteria in the control strategy of PCMR-001 back to the step plans and step reports. It found that none of the criteria could be traced to a plan as it was approved before execution. It also found that the master plan, PCMP-001, gave no rule from which the criteria could have been derived in advance. QA opened INV-001 the same day.

Version 1.0 reported 27 studies, executed between 2025-08-11 and 2026-05-15. Each of them closed before the Version 1.0 reports were approved. The count covers every study that started before this investigation closed, except the two augmentation studies of INV-002 (§2), which belong to that investigation, and five others. The comparability study S-CEX-FRC and the retests of retained samples by nrCE-SDS (S-PA-RET and S-VI-RET) produced only fragment results, which Version 1.0 did not report (§4.7). The further Protein A scale-down runs, study S-PA-SDM2, were made during the investigation (§2). The osmolality study S-BRX-OSM at the production bioreactor started on 2026-09-01, during the investigation, and is outside its scope. The bioreactor report PCR-003 Version 2.0 records it as PD-003-01, a discrepancy from the plan.

Two immediate actions were taken on the day the investigation was opened.

1. The Version 1.0 documents were placed on hold for BLA use. They were kept unchanged, as the record under investigation.
2. No analysis that depended on the Version 1.0 step criteria was to be issued until the investigation had concluded.

The investigation was closed on 2026-09-18.

2 Scope

The investigation covers the 20 Version 1.0 documents of the A-Mab characterization package, listed in Table 2.1. Within them it covers four subjects. These are the step acceptance criteria used in the proven acceptable range analyses, with their derivation and their place in the protocols, the method used for fragments, the capability assessment of the drug substance, and the execution records of the studies.

The adequacy of the response-surface models is outside this investigation. During it, the re-analysis found response-surface models that do not meet the model-adequacy criteria of PCMP-001 Version 2.0, and QA opened INV-002 on 2026-08-31 to investigate that finding. Under INV-002 corrective action CA3, adopted on 2026-09-15, before this investigation closed, every response-surface output is analysed by model selection, and INV-002 gives the procedure. The verification of the platform worst cases in §6.2, on which the Version 2.0 criteria rest, used these selected models. Every comparison of ranges in this report uses the models Version 1.0 fitted, so that it shows what the change of criteria moved. The step reports give the ranges on the selected models.

Table 2.1: Documents in the scope of the investigation, all at Version 1.0 when it was opened.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)

Document ID	Document class	Subject
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

Three things are outside the scope. The computation of a proven acceptable range itself, which is how the other parameters are held or varied while one parameter is scanned, is not reviewed here. The experiments themselves are not reviewed either. Finally, the study events recorded in the step reports keep the outcome those reports give them.

The re-analysis in §6 uses the same designed experiments and the same results as Version 1.0, with three additions. The first is the fragment results by nrCE-SDS of the designed experiments. They were measured before Version 1.0 was approved (§4.7), in the cation exchange response surface study, in the re-executed anion exchange design, in the virus filtration design and in the retests of retained samples (S-PA-RET and S-VI-RET), but Version 1.0 did not report them. The second is 6 further set-point runs of the Protein A scale-down model (study S-PA-SDM2), made from 2026-09-07 to 2026-09-11, during the investigation. The third is the UF/DF univariate study S-UFDF-UNI2. Only its aggregate results enter the re-analysis, in the verification of the platform worst cases (§6.2).

The Protein A plan, PCP-005, had planned 3 set-point runs for the qualification of that model (study S-PA-SDM). The further runs were made during the re-analysis, to narrow the confidence interval of the offset of the model for pool host cell protein. Of the Protein A pool attributes, host cell protein comes nearest its criterion of 16,719 ng/mg. On the model that PCR-005 Version 2.0 uses, its upper 95 % prediction limit over the NOR as characterized is 94 % of the criterion, and it rises with lower elution buffer pH and higher protein load. That model is fitted on the log scale, so the offset is a difference of mean logarithms. It is stated as a ratio, the geometric mean of the scale-down runs divided by that of the manufacturing-scale batches. On the planned runs alone, the ratio is 0.935, with a 90 % confidence interval of 0.795 to 1.100. On

all runs it is 0.973, with an interval of 0.891 to 1.063. The plan did not foresee the further runs, so the Protein A report PCR-005 Version 2.0 records them as PD-005-01, a discrepancy from the plan. They are not a deviation, because no approved procedure or GMP requirement was departed from.

A scale-down model is qualified for an output when the conclusion against its criterion over the NOR is the same with the offset at either end of that interval (§6.1). Over the NOR, the model is not qualified for pool host cell protein, on the planned runs alone or on all runs. The conclusion holds with the ratio at its estimate. With the ratio at the low end of its interval, which raises the prediction at manufacturing scale, the upper prediction limit over the NOR is 17,143 ng/mg on all runs, above the criterion. NORs come from production, and an offset does not narrow them. The Protein A report PCR-005 Version 2.0 therefore keeps the NOR in force, states that the model is not qualified for pool host cell protein over it, and flags the NOR for discussion with manufacturing. The further runs narrowed the interval but did not change the conclusion. The decision to make them was taken after the result on the planned runs was known. The qualification rule and the criterion were not changed, and only the number of runs grew. For leached Protein A the model is qualified on either set of runs.

3 Background

3.1 The master plan

PCMP-001 Version 1.0, the master plan, required that each step be judged against a criterion that applies to it. It did not say how that criterion was to be obtained. Its §7 left the threshold to the step plans.

> Where a threshold depends on the step, on the attribute or on the precision of the method, the criterion is defined in the plan of that step and the reason is given there.

The same section stated that the criterion for a response is the in-process limit of the step where one has been established, and the drug-substance criterion where none has. It gave neither a rule nor a value for an in-process limit.

3.2 The Protein A analysis

The need for step criteria became apparent during the analysis of the Protein A study. The capture pool carries host cell protein at a level far above the drug-substance criterion of 100 ng/mg. Protein A capture removes most of the host cell protein of the harvest, and part of the remainder is carried into the eluate bound to the antibody. That level is acceptable in an intermediate, because cation exchange and anion exchange remove most of it afterwards. Judged against the drug-substance criterion, no part of the Protein A characterization range would have been acceptable. PCMR-001 Version 1.0 put the argument in one sentence.

> The drug substance limit is the wrong yardstick for an intermediate, and the campaign uses a criterion for each step that applies to that step.

3.3 The Version 1.0 criteria

Version 1.0 set 15 step criteria at 6 steps, by four rules. Table 3.1 gives each criterion with its rule code, beside the Version 2.0 criterion that replaced it. The Version 2.0 columns are discussed in §6.

1. For an impurity that downstream steps clear, the drug-substance criterion was carried back through the clearance the downstream steps delivered at their setpoints in this campaign. The result was divided by a safety factor (R1, 6 criteria).

2. For an attribute that no downstream step changes, the criterion was an alert limit at the capability mean plus 2.5 SD (R2, 4 criteria, all at the bioreactor).
3. For viral clearance at low-pH viral inactivation and anion exchange, the criterion was the cumulative requirement less the nominal clearance of the other credited steps (R3, 3 criteria).
4. For viral clearance at virus filtration, the criterion was the nominal clearance claimed for the step less 0.35 log10 for assay variability (R4, 2 criteria).

Table 3.1: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0.

Step	Attribute	Version 1.0[a]		Version 2.0	
		Criterion	Rule	Criterion	Basis
Production bioreactor	Acidic variants (%)	≤ 25.8	R2	18.0–39.0	Carried back
	Afucosylation (%)	2.00–9.43	R2	2.00–13.0	Drug substance
	Aggregates, HMW (%)	≤ 1.86	R1, margin 6.8	≤ 6.30	Carried back, tightened[b] and widened[c] from ≤ 9.30
	Galactosylation (%)	10.0–30.1	R2	10.0–40.0	Drug substance
	High mannose (%)	3.00–7.28	R2	3.00–10.0	Drug substance
Protein A	Leached Protein A (ppm)	none[d]	–	≤ 158	Carried back
	Fragments, LMW (%)	not assessed[e]	–	≤ 2.45	Carried back
	Host cell protein (ng/mg)	≤ 10,724	R1, margin 4.5	≤ 16,719	Carried back, widened[c] from ≤ 16,000
Low-pH viral inactivation	Acidic variants (%)	none[d]	–	18.0–40.0	Drug substance
	Aggregates, HMW (%)	≤ 2.17	R1, margin 6.0	≤ 6.84	Carried back, tightened[b] from ≤ 10.0
	Fragments, LMW (%)	not assessed[e]	–	≤ 2.60	Carried back
	XMuLV clearance (log10)	≥ 4.93	R3	≥ 6.60	Platform claim
Cation exchange	Aggregates, HMW (%)	≤ 1.00	R1, margin 5.0	≤ 4.90	Carried back, tightened[b] from ≤ 5.00
	Host cell protein (ng/mg)	≤ 193	R1, margin 3.2	≤ 418	Carried back, widened[c] from ≤ 400
	Fragments, LMW (%)	not assessed[e]	–	≤ 2.00	Drug substance
Anion exchange	Host cell protein (ng/mg)	≤ 21.7	R1, margin 4.6	≤ 100	Drug substance
	Fragments, LMW (%)	not assessed[e]	–	≤ 2.00	Drug substance
	MVM clearance (log10)	≥ 3.28	R3	≥ 4.00	Platform claim
	XMuLV clearance (log10)	≥ 3.78	R3	≥ 5.50	Platform claim
Virus filtration	Fragments, LMW (%)	not assessed[e]	–	≤ 2.00	Drug substance
	MVM clearance (log10)	≥ 4.97	R4	≥ 4.62	Platform claim
	XMuLV clearance (log10)	≥ 5.47	R4	≥ 4.62	Platform claim

[a] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[b] Tightened on verification: a downstream step did worse within its NOR than the platform worst case.
[c] Widened on verification: a downstream step did better than the platform worst case at the one-sided 95 % prediction bound of a single batch, on a model that meets the first three adequacy criteria of PCMP-001 §7.2.
[d] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[e] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
[R1] Version 1.0 rule: nominal back-calculation / margin.
[R2] Version 1.0 rule: mean + 2.5 sd of capability.
[R3] Version 1.0 rule: requirement − nominal of other steps.
[R4] Version 1.0 rule: nominal claim less 0.35 log10 for assay variability.

The back-calculation through nominal downstream clearance, the viral criteria less the other steps’ nominal clearance and the virus filtration criterion less the assay allowance all rest on the nominal results of this campaign. The alert limits rest on its capability data. Every Version 1.0 criterion was therefore set after the data were seen, from this campaign’s nominal performance or capability, and where a safety factor was used its size was chosen step by step.

4 Investigation

4.1 Approach

The sequence in which the criteria were set was reconstructed from the analysis records, from the study chronology and from the authors of the Version 1.0 reports. Each criterion in Table 3.1 was recomputed from its stated rule and compared with the value printed in the documents. The capability assessment was repeated against the drug-substance criteria as the register prints them. The execution windows of the studies were compared with the calibration records of the equipment they used. The record of the fragment method was reviewed against the decision AMD-001 to measure fragments by nrCE-SDS and against the comparability study S-CEX-FRC. Every recomputed criterion agreed with the printed value.

4.2 F1. Criteria defined after the data were seen

The step criteria were defined after the data were seen and outside the approved protocols.

Each step plan was approved before its studies began, as Table 4.3 in §4.6 shows. The Protein A plan PCP-005, for example, was approved on 2025-11-03, and its first study started on 2025-11-05. None of the step criteria was in a plan as approved. The criteria were introduced later, while the Protein A data were being analysed, when it became clear that the capture pool could not be judged against the drug-substance criterion (§3.2). They were then written into the text of every step plan when the Version 1.0 package was compiled. The approved plans were not revised to carry them. The Version 1.0 plans therefore read as if the criteria had been approved before execution.

The master plan PCMP-001, approved on 2025-08-04, did not prevent this. As §3.1 shows, it required a step criterion without fixing one. The record therefore holds no protocol, approved before execution, that contains a step criterion. A criterion added to a protocol after its data are known does not meet the purpose of a protocol (U.S. Food and Drug Administration 2011).

4.3 F2. Safety factors chosen so that each step passed at its set-point

For an impurity that downstream steps clear, Version 1.0 carried each drug-substance criterion back through the nominal downstream clearance of this campaign and then divided it by a safety factor (R1 in Table 3.1). The factors ranged from 3.2 to 6.8, and no document records why they differ.

Table 4.1 sets each factor against the value that the response surface model of the step predicts at its set-point. The last column gives the largest factor at which that set-point value would still have met the criterion. Every step meets its Version 1.0 criterion at its own set-point. Had the largest factor used, 6.8, been applied to every step, the set-point value would have failed its criterion for 4 of the attributes in Table 4.1. The analysis records show that a factor was reduced where the limit would otherwise have fallen below the value the step produces at its set-point.

Table 4.1: Safety factors of the Version 1.0 step criteria against the value each step produces at its set-point.

Step	Attribute	Highest pool level[a)]	Safety factor	Version 1.0 criterion	Predicted at set-point	Largest factor met at set-point
Production bioreactor	Aggregates, HMW (%)	12.7	6.8	≤ 1.86	1.52	8.32
Protein A	Host cell protein (ng/mg)	48,260	4.5	≤ 10,724	7,216	6.69
Low-pH viral inactivation	Aggregates, HMW (%)	13.0	6.0	≤ 2.17	1.91	6.81
Cation exchange	Aggregates, HMW (%)	5.00	5.0	≤ 1.00	0.75	6.66
	Host cell protein (ng/mg)	619	3.2	≤ 193	141	4.38
Anion exchange	Host cell protein (ng/mg)	100.0	4.6	≤ 21.7	14.8	6.75

[a)] The highest pool level that the nominal clearance of the downstream steps brings within the drug-substance criterion. Nominal clearance is the clearance each downstream step showed at its set-point in this campaign.

The highest pool level in Table 4.1 assumes that every downstream step clears as much as it does at its set-point. A downstream step that runs elsewhere in its NOR may clear less, and the downstream steps can then bring a lower pool level within the drug-substance criterion. Version 1.0 divided each criterion by a safety factor to cover this. The factor was chosen. It was not calculated from how much less the downstream steps clear within their NORs.

The Version 1.0 viral clearance criteria for low-pH viral inactivation and anion exchange rest on nominal clearance in the same way. Each was the cumulative requirement less the clearance of the other two steps in the viral clearance summary of Version 1.0. That summary gives a single value for each step and virus, at the set-points. For XMuLV it gives 7.10 log10 for low-pH viral inactivation, 5.95 for anion exchange

and 5.82 for virus filtration. For MVM it gives 4.71 log10 for anion exchange and 5.32 for virus filtration, and no clearance for low-pH viral inactivation. The Version 1.0 XMuLV criterion for low-pH viral inactivation, for example, was the cumulative requirement of 16.7 log10 less 5.95 and 5.82, which is 4.93 log10. The Version 1.0 virus filtration criteria were the virus filtration values of the same summary less 0.35 log10. None of these summary values is a mean of the response surface models in Table 6.5.

4.4 F3. Capability limits judged against the data they came from

For the glycans and acidic variants, Version 1.0 set 4 bioreactor criteria at the mean plus 2.5 SD of the commercial-scale capability simulation (R2 in Table 3.1). Table 4.2 recomputes each one. The recomputed value and the printed criterion agree to the second decimal place.

Table 4.2: Version 1.0 bioreactor criteria recomputed from the capability data. Mean and SD are those of the simulated drug substance at commercial scale.

Attribute	Capability mean	Capability SD	Mean + 2.5 SD	Version 1.0 upper criterion	Drug-substance criterion
Acidic variants (%)	22.30	1.386	25.76	25.76	18.0–40.0
Afucosylation (%)	7.98	0.580	9.43	9.43	2.00–13.0
Galactosylation (%)	25.98	1.631	30.06	30.06	10.0–40.0
High mannose (%)	5.99	0.515	7.28	7.28	3.00–10.0

The capability simulation draws every parameter inside its NOR. The proven acceptable range analysis then asked whether the upper 95 % prediction limit stayed within the alert limit while the other parameters varied inside the same NOR. For a normal distribution, about 0.62 % of the batches lie above a one-sided limit at the mean plus 2.5 SD, whatever the process. A limit of this kind is an alert limit for trending. It is not an acceptance criterion, because it carries no information about what the product needs.

An alert limit of this kind can be reached inside the NOR by a batch that is well within the drug-substance criterion. The bioreactor report PCR-003 Version 1.0 reported such a case, a corner of the NOR at which predicted afucosylation exceeded its alert limit while meeting the drug-substance criterion by a wide margin. For the same reason, several proven acceptable ranges at the bioreactor stopped short of the NOR because they reached an alert limit, not the drug-substance criterion (§6.3).

4.5 F4. Acidic variants assessed against one end of a two-sided range

The register gives acidic variants a two-sided criterion of 18 % to 40 %, and the Version 1.0 capability table printed that range. The capability index beside it was computed against the upper end alone and printed as 4.3. The simulated drug substance has a mean of 22.3 % and an SD of 1.39 %. Computed against both ends, the index is 1.03, because the mean lies 3.1 SD above the lower limit and 12.8 SD below the upper limit. The same one-sided treatment was used in the proven acceptable range analysis. Table 3.1 records the Version 1.0 alert limit for acidic variants as an upper limit only, and PCR-003 Version 1.0 judged its acidic variant ranges against that upper limit alone.

Two statements in PCMR-001 Version 1.0 rested on the one-sided capability index. It reported the lowest capability index of the drug substance as 1.51, for MVM clearance. A capability index for a validated clearance claim, computed per batch, has no clear meaning, and it is quoted here only as PCMR-001 printed it. PCMR-001 also reported that no simulated batch fell outside an acceptance criterion. Against both ends of the acidic variant range, the simulation puts one batch in 2,000 outside it.

4.6 F5. Approval and execution dates missing from the records

The Version 1.0 records misstated when the plans were approved and did not state when the studies ran. Every Version 1.0 title block carried the date 2026-07-24. That was the approval date of the reports and the date on which the package was compiled. For the plans, it replaced the date of approval. Table 4.3 sets the approval date of each plan against the start of its first study. Every plan was approved before its first study began. Read from the title blocks, every plan appeared to have been approved after its studies had started.

Table 4.3: Approval date of each step plan, against the start of its first study. The Version 1.0 title block of every plan printed 2026-07-24 instead of the approval date.

Plan	Step	Approved	First study started
PCP-003	Production bioreactor	2025-08-07	2025-08-11
PCP-004	Harvest	2025-10-20	2025-11-03
PCP-005	Protein A	2025-11-03	2025-11-05
PCP-006	Low-pH viral inactivation	2026-01-05	2026-01-07
PCP-007	Cation exchange	2026-02-02	2026-02-04
PCP-008	Anion exchange	2026-02-02	2026-02-04
PCP-009	Virus filtration	2026-04-13	2026-04-15
PCP-010	UF/DF	2026-02-23	2026-02-25

The Version 1.0 reports gave no execution date for any study. Their equipment tables recorded each instrumented system as in calibration at execution and gave the next calibration due date. Every one of these due dates precedes the date in the title blocks. A reader of the record could not confirm calibration at the time of use, and the tables appeared to cite expired calibrations.

The execution windows were recovered from the study records. Table 4.4 gives them for the 7 studies that ran on the 3 instrumented systems. Every one of these studies closed before the calibration of its system fell due. The smallest margin is at study S-UFDF-UNI at the UF/DF step, on the ultrafiltration / diafiltration TFF skid EQ-TFF-142. It closed on 2026-03-27, 26 days before the calibration due on 2026-04-22.

Table 4.4: Execution windows of the studies that used instrumented systems, against the calibration due date and the date in the Version 1.0 title blocks.

Equipment	Description	Next calibration due	Study executed on the system			Date in Version 1.0 title block
			Study	Start	End	
EQ-CHR-118	temperature-controlled chromatography chamber	2026-05-30	S-CEX-RSM	2026-03-02	2026-03-27	2026-07-24
EQ-BRX-205	production scale-down bioreactor pCO2 probe	2026-06-14	S-BRX-SDM S-BRX-SCR S-BRX-UNI S-BRX-RSM	2025-08-11 2025-09-08 2025-10-06 2025-10-20	2025-09-05 2025-10-03 2025-10-31 2025-12-12	2026-07-24
EQ-TFF-142	ultrafiltration / diafiltration TFF skid	2026-04-22	S-UFDF-SDM S-UFDF-UNI	2026-02-25 2026-03-09	2026-03-06 2026-03-27	2026-07-24

4.7 F6. Fragments by SEC in the Protein A and low-pH viral inactivation pools

In the comparability study S-CEX-FRC at cation exchange, executed from 2026-03-02 to 2026-03-13, fragments were measured by SEC and by nrCE-SDS on the same samples. Table 4.5 gives every replicate.

Table 4.5: Fragments by SEC and by non-reduced CE-SDS on the same samples (study S-CEX-FRC).

Sample	Replicate	Fragments, LMW (%)		SEC minus nrCE-SDS
		SEC	nrCE-SDS	
Protein A pool	1	1.32	0.90	+0.42
	2	1.42	1.10	+0.32
	3	1.32	0.82	+0.50
	4	1.29	0.83	+0.46
VI pool	1	1.43	0.84	+0.59
	2	1.39	0.88	+0.51
	3	1.42	0.88	+0.54
	4	1.49	1.02	+0.47
CEX load	1	1.36	0.85	+0.52
	2	1.52	1.03	+0.49
CEX fraction F2	1	1.65	1.56	+0.09
CEX fraction F3	1	0.64	0.76	-0.11
CEX fraction F4	1	0.40	0.49	-0.09
CEX fraction F5	1	0.42	0.32	+0.10
CEX pool	1	0.61	0.62	-0.01
	2	0.70	0.57	+0.13
	3	0.51	0.61	-0.10
	4	0.70	0.72	-0.02
AEX pool	1	0.55	0.65	-0.10
	2	0.68	0.70	-0.02

nrCE-SDS reports the sum of clipped species. The partial molecules seen under SDS (HHL, HH, HL, free light chain), which come from antibody lacking one or more interchain disulfide bonds, are excluded. This makes the two methods comparable.

On average, SEC read fragments higher than nrCE-SDS by 0.43 to 0.53 percentage points in the Protein A pool, the VI pool and the CEX load. From the CEX pool onward, the mean difference between the two methods was no more than 0.06 percentage points. The difference follows the host cell protein content of the sample. Part of the host cell protein elutes in the low molecular weight window of SEC, where it is counted with the clipped antibody species. The excess can therefore be no larger than the host cell protein in the sample. Cation exchange removes most of the host cell protein, and from its pool onward the excess is close to zero. In the Protein A pool, SEC reported a mean of 1.34 % fragments, where nrCE-SDS reported 0.91 %. The excess there is about 47 % of the fragment level that nrCE-SDS measures, and about 17 % of the Protein A fragment criterion of 2.45 % (Table 6.1). SEC fragment results from the Protein A pool and the low-pH viral inactivation pool are therefore biased high by an amount that would use a large part of a fragment criterion.

The decision AMD-001 to measure fragments by nrCE-SDS was taken on 2026-03-20 on these data. From that date, fragments were measured by nrCE-SDS in every study,

at every step, so that one method spans the train. The CEX response surface study, which ran from 2026-03-02 to 2026-03-27 and so spanned the decision, measured fragments by both methods in parallel. The retained response surface samples of Protein A and viral inactivation were retested by nrCE-SDS from 2026-04-06 to 2026-04-17. The screening samples had not been retained. Their SEC results are kept and marked as not used. The re-executed anion exchange design (DEV-008-01) started on 2026-03-23, after the decision, and measured fragments by nrCE-SDS. Its invalidated first execution had used SEC. Fragments were not in the scope of the bioreactor design. Table 4.6 shows which method ran in which study.

Table 4.6: Methods used for size variants in the designed studies, by step.

Step	Attribute	Method used in study	
		Screening	Response surface
Production bioreactor	Aggregates, HMW	SEC	
	Fragments, LMW	not in scope[a)]	
Harvest	Aggregates, HMW	no DoE[b)]	
	Fragments, LMW	no DoE[b)]	
Protein A	Aggregates, HMW	SEC	
	Fragments, LMW	SEC[c)]	SEC[c)], nrCE-SDS[d)]
Low-pH viral inactivation	Aggregates, HMW	SEC	
	Fragments, LMW	SEC[c)]	SEC[c)], nrCE-SDS[d)]
Cation exchange	Aggregates, HMW	SEC	
	Fragments, LMW	SEC	SEC, nrCE-SDS[e)]
Anion exchange	Aggregates, HMW	not measured[f)]	
	Fragments, LMW	nrCE-SDS[g)]	nrCE-SDS[g)]
Virus filtration	Aggregates, HMW	not measured[f)]	
	Fragments, LMW	nrCE-SDS	nrCE-SDS
UF/DF	Aggregates, HMW	no DoE[b)]	
	Fragments, LMW	no DoE[b)]	

[a)] Fragments were not an attribute of this study.
[b)] Univariate characterization only.
[c)] In the Protein A and low-pH viral inactivation pools part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[d)] Retained samples of the response-surface design, tested 2026-04-06 to 2026-04-17.
[e)] Both methods on the same samples of this design.
[f)] Aggregate was not measured in the designs at this step.
[g)] Re-executed design (DEV-008-01). The invalidated first execution measured fragments by SEC.

The decision AMD-001 and the nrCE-SDS retests predate Version 1.0. However, the Version 1.0 documents did not report fragments, their register carried no fragment attribute, and none of them cited AMD-001. No procedure required that a method decision taken during a campaign be assessed against the approved plans and, where it changed them, be put into effect by a plan amendment. The decision to measure fragments by nrCE-SDS therefore never reached the documents.

5 Root cause

The root cause is that PCMP-001 Version 1.0, the master plan, required a criterion for each step but gave no rule for deriving one (finding F1). When the Protein A analysis showed that an intermediate cannot be judged against a drug-substance criterion, the criteria were derived during the analysis, by the people analysing the data, with the results in view. Every rule they chose drew on the nominal results or the capability data of the studies being judged (§3.3).

Four causes contributed.

1. SOP-1002 did not require that the acceptance criteria and their basis be approved in the protocol before execution. Criteria set after execution could therefore be added to the plan text when the package was compiled, without a revision of the approved plan (finding F1, criteria defined after the data were seen).
2. Neither the plans nor the master plan defined how much less a downstream step may clear within its NOR, which is what a step criterion has to allow for. A chosen safety factor was used instead, and its size was set step by step (finding F2). Capability data were used where no downstream step changes the attribute (finding F3).
3. No second review checked whether a capability calculation used a one-sided or a two-sided criterion as the register gives it (finding F4, acidic variants). No second review checked the dates in the title blocks and study records against the approval and equipment records (finding F5).
4. No procedure required that a method decision be assessed against the approved plans and put into effect by plan amendment (finding F6, fragments by SEC).

6 Impact assessment

6.1 The Version 2.0 criteria

Under corrective action CA1, the investigation defined the intermediate acceptance criteria and the rules for judging against them. Corrective action CA1 then issues them in PCMP-001 Version 2.0, the master plan, effective 2026-09-30. The investigation closed on 2026-09-18, before that date, and the re-analysis in this section applied the rules as CA1 defined them.

Each drug-substance criterion is carried backward through the train, using the worst case each downstream step showed on the platform (X-Mab, Y-Mab and Z-Mab). The platform worst case is the lowest fold clearance or the largest increase seen on those products. No margin is applied. The glycans keep their drug-substance criterion, because no downstream step changes them. Each viral clearance step is held to its platform claim. The claims sum to the cumulative clearance that the CQA register requires of the process, taken from the A-Mab case study (CMC Biotech Working Group 2009). The sums are 16.72 against a requirement of 16.7 log10 for XMuLV and 8.62 against 8.6 log10 for MVM.

Corrective action CA1 also defined three rules that the re-analysis applies.

1. Verification compares each platform worst case with the A-Mab worst case that the response surface model of the downstream step predicts within its NOR. The bound that can replace a platform value is the one-sided 95 % prediction bound of a single batch at the least favourable combination of NOR low, set-point and NOR high (the single-batch bound). Where the A-Mab prediction is worse than the platform value, the single-batch bound replaces it and the criteria upstream tighten. Where the single-batch bound is better than the platform value, it replaces it and the criteria upstream widen, but only if the model passes the adequacy tests of PCMP-001 Version 2.0 (R^2 less predicted R^2 at most 0.2, and no lack of fit at the 0.05 level). Otherwise the platform value stands. No criterion is set from a point prediction or from a confidence bound of the mean.
2. A scale-down model is qualified for an output when its offset (the mean of the scale-down set-point runs less the mean of the manufacturing-scale batches) is known precisely enough that the conclusion over the NOR is the same at either end of the 90 % confidence interval of the offset. A qualified model may differ in mean from manufacturing scale. Every Version 2.0 judgement subtracts the offset, so results are stated at manufacturing scale.
3. A quality attribute is judged per batch, on the 95 % prediction interval of a single batch. Viral clearance is judged on the mean predicted log reduction factor (LRF)

at the least favourable combination of NOR low, set-point and NOR high of the other parameters.

Table 6.1 gives the 22 criteria with their derivation. Of these, 8 are the drug-substance criterion applied directly, 9 are carried back through the downstream steps, and 5 are platform viral clearance claims. When the platform worst cases were verified (§6.2), 3 of the back-calculated criteria were tightened and 3 were widened. The criterion for aggregates at the production bioreactor step is counted in both, because two downstream steps tightened it and another widened it.

Table 6.1: Intermediate acceptance criteria and their derivation from the drug-substance criteria.

Attribute	DS criterion	Step output	Downstream worst case (platform or verified)		IAC	Basis
			Clearance factor	Increase		
Host cell protein (ng/mg)	≤ 100	Protein A	167	0.00	≤ 16,719	Carried back, widened[a]
		Cation exchange	4.18	0.00	≤ 418	Carried back, widened[a]
		Anion exchange	1.00	0.00	≤ 100	DS criterion[b]
Aggregates, HMW (%)	≤ 5.00	Production bioreactor	1.40	0.64	≤ 6.30	Carried back, tightened[c] and widened[a]
		Low-pH viral inactivation	1.40	0.10	≤ 6.84	Carried back, tightened[c]
		Cation exchange	1.00	0.10	≤ 4.90	Carried back, tightened[c]
Fragments, LMW (%)	≤ 2.00	Protein A	1.30	0.15	≤ 2.45	Carried back
		Low-pH viral inactivation	1.30	0.00	≤ 2.60	Carried back
		Cation exchange	1.00	0.00	≤ 2.00	DS criterion[b]
		Anion exchange	1.00	0.00	≤ 2.00	DS criterion[b]
		Virus filtration	1.00	0.00	≤ 2.00	DS criterion[b]
Leached Protein A (ppm)	≤ 5.00	Protein A	31.6	0.00	≤ 158	Carried back
Acidic variants (%)	18.0–40.0	Production bioreactor	1.00	1.00	18.0–39.0	Carried back
		Low-pH viral inactivation	1.00	0.00	18.0–40.0	DS criterion[b]
Afucosylation (%)	2.00–13.0	Production bioreactor	1.00	0.00	2.00–13.0	DS criterion[b]
Galactosylation (%)	10.0–40.0	Production bioreactor	1.00	0.00	10.0–40.0	DS criterion[b]
High mannose (%)	3.00–10.0	Production bioreactor	1.00	0.00	3.00–10.0	DS criterion[b]
XMuLV clearance (log10)	≥ 16.7[d]	Low-pH viral inactivation	not applicable		≥ 6.60	Platform claim[e]
		Anion exchange	not applicable		≥ 5.50	Platform claim[e]
		Virus filtration	not applicable		≥ 4.62	Platform claim[e]
MVM clearance (log10)	≥ 8.60[d]	Anion exchange	not applicable		≥ 4.00	Platform claim[e]
		Virus filtration	not applicable		≥ 4.62	Platform claim[e]

[a] The one-sided 95 % prediction bound of a single batch at a downstream step's A-Mab worst case within its NOR was better than the platform worst case. The bound replaced it.
[b] No downstream step changes the attribute, so the drug-substance criterion applies.
[c] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The one-sided 95 % prediction bound of a single batch there replaced it.
[d] Cumulative requirement across the steps that clear the virus.
[e] Log reduction claimed for this step in the modular viral clearance claim.
DS, drug substance. IAC, intermediate acceptance criterion. Increase, in the unit of the criterion.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

This campaign’s data enter the Version 2.0 criteria, as they entered the Version 1.0 criteria. The difference lies in how they enter. Version 1.0 set its criteria after the data were seen, from the nominal performance of the downstream steps, with a safety factor chosen step by step. Version 2.0 starts from the platform worst case, by a rule written before any report applied it. This campaign’s data can then move a criterion only through the single-batch bound of a downstream step within its NOR, in either direction. The data of a step never set its own criterion.

6.2 Verification of the platform worst cases

The platform worst cases are assumptions about A-Mab. Each was checked against the data of the downstream step it describes, under the first rule of CA1 in §6.1. Each prediction of the downstream step is converted to manufacturing scale with the offset of its scale-down model before the fold or the increase is computed, under the second rule of CA1. The value observed at the edges of the characterization range is shown for information. Table 6.2 gives the result. Of the 8 assumptions that could be tested, 6 were confirmed, because the prediction within the NOR was no worse than the platform value. For 2 of them the single-batch bound was also better than the platform value, on a model that meets the first three adequacy criteria of PCMP-001 §7.2, and the criteria upstream were recalculated with the bound. For the increase of acidic variants at low-pH viral inactivation the bound was also better, but the acidic-variant results of that design have no scatter. The model fits them exactly, so the lack-of-fit test and the model F-test cannot be applied, and the platform value stands. For the other three the single-batch bound was not better than the platform value, so the platform value stands.

Table 6.2: Platform worst cases assumed by the intermediate acceptance criteria, against this campaign's results.

Step	Attribute	Worst case	Platform	This campaign			Outcome
				At the edges	Within the NOR	95 % bound	
Low-pH viral inactivation	Aggregates, HMW	Increase, largest (points)	0.70	0.82	0.50	0.54	Better, widened[a)]
	Fragments, LMW		0.15	0.22	0.10	0.19	Confirmed
	Acidic variants		1.00	0.33	0.13	0.13	Confirmed, not widened[b)]
Cation exchange	Host cell protein	Clearance factor, lowest	40.0	27.2	40.2	32.9	Confirmed
	Aggregates, HMW		2.00	1.00	1.58	1.40	Not confirmed, tightened[c)]
	Fragments, LMW		1.30	1.28	1.45	1.27	Confirmed
	Leached Protein A		3.16	not a response[d)]			
Anion exchange	Host cell protein	Clearance factor, lowest	4.00	3.55	5.40	4.18	Better, widened[a)]
	Leached Protein A		10.0	not a response[d)]			
UF/DF	Aggregates, HMW	Increase, largest (points)	0.00	0.07	0.06	0.10	Not confirmed, tightened[c)]

[a)] The one-sided 95 % prediction bound of a single batch is better than the platform worst case. The criteria upstream of this step were recalculated with the bound.
[b)] The one-sided 95 % prediction bound of a single batch is better than the platform worst case, but the model is not adequate by the criteria of PCMP-001: it fails a test, or it fits the results exactly and the tests cannot be applied. The platform value stands.
[c)] The criteria upstream of this step were recalculated with the one-sided 95 % prediction bound of a single batch at the value predicted within the NOR.
[d)] Not a response of the response-surface study at this step, so the platform value stands unverified.
Computed from unrounded values. A criterion recomputed from the rounded values shown can differ in its last digit.

At the edges of the characterization range, cation exchange cleared host cell protein 27.2-fold, short of the platform worst case of 40.0-fold. Within the NOR, the lowest clearance the model predicts is 40.2-fold, and the single-batch bound is 32.9-fold. The edges of the characterization range lie outside the NOR, so the edge value does not decide the criterion. The prediction within the NOR confirms the platform value. The single-batch bound is below it, so the platform value of 40.0-fold stands for cation exchange. The verification used the models that INV-002 corrective action CA3 selects for each output (§2). At anion exchange, the single-batch bound for host cell protein is 4.18-fold, which is better than the platform value of 4.00-fold. The anion exchange model passes the adequacy tests, so this bound replaced the platform value. The anion exchange bound widened the cation exchange criterion for host cell protein from 400 to 418 ng/mg. Carried back through the platform value at cation exchange and this bound at anion exchange, the Protein A criterion for host cell protein was widened from 16,000 to 16,719 ng/mg. The Protein A criterion is also wider than the Version 1.0 limit of 10,724 ng/mg, which was never an approved criterion (finding F1). In all, 3 criteria were widened, for aggregates at the production bioreactor step,

host cell protein at the Protein A step and host cell protein at the cation exchange step. Table 6.3 in §6.3 gives the ranges that widened as a result.

Two assumptions were not confirmed, both for aggregate. The platform worst case for aggregate at cation exchange is a fold clearance of 2.0. Within its NOR, the lowest fold clearance the response surface model of this campaign predicts is 1.58, and the single-batch bound is 1.40. PCMP-001 assumed that no step after cation exchange changes aggregate. UF/DF raises it within its NOR, and the single-batch bound of the increase is 0.10 percentage points. The aggregate criteria upstream were recalculated with both bounds, and the 3 aggregate criteria at the production bioreactor, low-pH viral inactivation and cation exchange steps were tightened. The cation exchange criterion moved on the UF/DF bound alone, from the drug-substance criterion of 5 % to 4.90 %. At the bioreactor the aggregate criterion was also widened, because the single-batch bound of the aggregate increase at low-pH viral inactivation, 0.54 percentage points, is smaller than the platform value of 0.70. The platform value is an increase over a hold of 180 min. The A-Mab value is an increase within the NOR as characterized, where the hold lasts at most 120 min. For aggregate, the largest increase measured at the edges of the range studied, which include a hold of 180 min, is 0.82 points, above the platform value. That criterion therefore moved because the A-Mab hold is shorter, not because A-Mab is more stable at low pH. The criterion is still right, because routine manufacture does not hold longer than the NOR. The acidic-variant increase at low-pH viral inactivation, 0.13 points, is also below its platform value of 1.00, but its model fits exactly, so the bioreactor criterion for acidic variants stays at its a priori value. Against the tightened criteria, every aggregate proven acceptable range at these steps still spans the whole characterization range, so the tightening moved no range.

The criterion of 158 ppm for leached Protein A at the Protein A step rests on two platform worst cases, for cation exchange and anion exchange, that no response of this campaign could test (not a response in Table 6.2). Together they assume a clearance of 31.6-fold. Judged per batch at manufacturing scale, the upper 95 % prediction limit for the Protein A pool is at most 5.87 ppm over the NOR (Table 6.4). It meets the step criterion. For a batch at that level to meet the drug-substance criterion of 5 ppm, the downstream steps need to clear only 1.17-fold. The untested platform worst cases therefore carry little of the weight of this criterion.

6.3 Impact by document

All 20 documents in Table 2.1 are to be reissued at Version 2.0, effective 2026-09-30. Corrective action CA1 reissues the master plan PCMP-001 with the criteria and rules of §6.1. Corrective action CA2 amends the step plans PCP-003 to PCP-010 to state the criteria with their basis. Corrective action CA3 reissues the step reports, the master report PCMR-001, the Process Transfer Plan PTP-001 and the Pre-Characterization Risk Assessment RA-001. The step reports judge each step against the criteria, on the models selected under INV-002 (§2), and PCMR-001 consolidates the result. Each Version 2.0 document will say what changed in it under “Changes in this revision” and refer to the findings by number.

Table 6.3 compares the proven acceptable ranges in each report with a DoE, as Version 1.0 reported them and as the re-analysis gives them. These are the ranges computed with the other parameters varying within their NORs. The reports without a DoE, the harvest report PCR-004 and the UF/DF report PCR-010, contain no such ranges. Their revision will record the execution windows (finding F5).

Table 6.3: Proven acceptable ranges with the other parameters varying within their NORs, in Version 1.0 and in the re-analysis, by report.

Report	Step	In Version 1.0	Unchanged	Wider	Narrower	Moved[a)]	No range[b)]	New[c)]
PCR-003	Production bioreactor	20	3	13	1	3	0	0
PCR-005	Protein A	8	3	5	0	0	0	4
PCR-006	Low-pH viral inactivation	9	5	1	0	0	3	3
PCR-007	Cation exchange	8	2	6	0	0	0	4
PCR-008	Anion exchange	12	8	2	2	0	0	4
PCR-009	Virus filtration	4	2	2	0	0	0	2

[a)] One end of the range moved inward and the other outward.
[b)] With the other parameters at their least favourable combination of NOR low, set-point and NOR high, the viral clearance claim is not met even with the scanned parameter at its set-point.
[c)] Fragment ranges measured by nrCE-SDS, which Version 1.0 did not report.

Of the 38 ranges that changed, 29 widened. Of these, 13 are at the bioreactor. There, 11 glycan ranges widened because the capability alert limits of Version 1.0 (finding F3) gave way to the drug-substance criteria. A further two aggregate ranges widened because the Version 1.0 aggregate limit, divided by a safety factor (finding F2), gave way to a back-calculated criterion with no margin. At the other steps, 13 of the 16 wider ranges belong to criteria that Version 1.0 had divided by a safety factor. The remaining wider ranges are one range for leached Protein A at the Protein A step, which Version 1.0 judged against the drug-substance criterion itself, and two ranges at virus filtration, where the Version 1.0 criterion (the fourth rule in §3.3) lay above the platform claim that replaced it.

At cation exchange, the Version 1.0 aggregate criterion of 1.00 % HMW was the drug-substance criterion divided by a safety factor of 5.0. Against it, the proven acceptable range for protein load, with the other parameters varying within their NORs, was 10–27.6 g/L resin. That range did not reach the upper end of the protein load NOR at 32 g/L resin. Against the Version 2.0 criterion of 4.90 %, the range is 10–40 g/L resin, and it covers the upper end of the NOR. At the bioreactor, the range for dissolved CO2 judged on galactosylation, computed the same way, was 74.5–160 mmHg at Version 1.0. It did not reach the lower end of the NOR at 70 mmHg. Judged against the drug-substance criterion, it is 40–160 mmHg.

The ranges that did not widen have three causes. At the bioreactor, the only Version 2.0 limit stricter than its Version 1.0 counterpart is the lower limit for acidic variants, which Version 1.0 did not apply (finding F4, §4.5). As a result, four of the acidic variant ranges there changed. Of these, one narrowed, three moved and none widened.

Judged against the upper limit alone, every one of them would span the whole characterization range, so the lower limit of 18 % now sets them. Acidic variants fall as CO2 partial pressure (pCO2) rises, in this campaign's response surface model and in the response surface estimates of the published A-Mab case study (CMC Biotech Working Group 2009, Table 3.16). For CO2 partial pressure (pCO2), the range is therefore 40–115 mmHg, and it no longer reaches the upper end of the NOR at 130 mmHg.

At low-pH viral inactivation, the XMuLV criterion rose from the Version 1.0 value of 4.93 log10 to the platform claim of 6.6 log10, and XMuLV clearance is now judged at the least favourable combination of the other parameters. The XMuLV clearance of low-pH viral inactivation therefore no longer gives a proven acceptable range for any of its three parameters. For inactivation pH, Version 1.0 had reported 3.2–3.96. At anion exchange, the viral clearance criteria also rose to the platform claims and are also judged at the least favourable combination. The load pH ranges there narrowed slightly, for XMuLV from 7.2–7.8 to 7.21–7.8 and MVM from 7.2–7.8 to 7.29–7.8.

Table 6.4 gives each pool attribute of each step against its Version 2.0 criterion over the NOR as characterized, at manufacturing scale. Each is judged on the 95 % prediction interval for a single batch, from the response surface model that Version 1.0 fitted (§2). Of the 17 pool attributes, 16 meet their criterion across the NOR.

For acidic variants at the production bioreactor step, the lower 95 % prediction limit for a single batch falls to 14.7 %, against the lower limit of 18.0 %. In this campaign's response surface model, higher culture pH and higher CO2 partial pressure (pCO2) lower acidic variants, and higher culture temperature raises them. The published A-Mab case study reports the same directions (CMC Biotech Working Group 2009, Table 3.16). The limit is therefore lowest at the corner of the NOR where culture pH and CO2 partial pressure (pCO2) are at their upper ends (6.95 and 130 mmHg) and culture temperature is at its lower end (34.5 °C). Culture duration is at its upper end (18 days) at that corner, but it has no significant effect on acidic variants in the model. With every parameter at its set-point the batch interval meets the criterion. The bioreactor report PCR-003 Version 2.0 reports the resulting range for CO2 partial pressure (pCO2), 40–115 mmHg.

Table 6.4: Pool attributes of each step against their Version 2.0 criteria, at manufacturing scale (the offset of the scale-down model is subtracted where one was determined). The mean and the 95 % prediction interval for a single batch come from the response surface model. The interval is taken over every combination of NOR low, set-point and NOR high of the parameters in the model.

Step	Attribute	Criterion	Mean at set-points	95 % prediction interval over the NOR	Met
Production bioreactor	Afucosylation (%)	2.00–13.0	8.42	5.43 to 11.2	Met
	Galactosylation (%)	10.0–40.0	26.0	18.6 to 35.2	Met
	High mannose (%)	3.00–10.0	6.34	4.55 to 8.45	Met
	Acidic variants (%)	18.0–39.0	22.6	14.7 to 29.9	Not met
	Aggregates, HMW (%)	≤ 6.30	1.65	1.11 to 2.20	Met
Protein A	Host cell protein (ng/mg)	≤ 16,719	7,467	1,729 to 15,872	Met
	Leached Protein A (ppm)	≤ 158	4.05	2.17 to 5.87	Met
	Fragments, LMW (%)	≤ 2.45	0.91	0.69 to 1.14	Met
Low-pH viral inactivation	Aggregates, HMW (%)	≤ 6.84	1.89	1.69 to 2.10	Met
	Acidic variants (%)	18.0–40.0	20.3	20.2 to 20.4	Met
	Fragments, LMW (%)	≤ 2.60	0.98	0.86 to 1.13	Met
Cation exchange	Aggregates, HMW (%)	≤ 4.90	0.76	0.36 to 1.45	Met
	Host cell protein (ng/mg)	≤ 418	159	73.4 to 267	Met
	Fragments, LMW (%)	≤ 2.00	0.62	0.44 to 0.81	Met
Anion exchange	Host cell protein (ng/mg)	≤ 100	15.3	3.27 to 28.9	Met
	Fragments, LMW (%)	≤ 2.00	0.61	0.40 to 0.82	Met
Virus filtration	Fragments, LMW (%)	≤ 2.00	0.64	0.35 to 0.86	Met

Table 6.5 gives the viral clearance of each step against its platform claim, on the same response surface models as Table 6.4 and at the NOR and set-points as characterized. With every parameter at its set-point, every one of the 5 claims is met. At the least favourable combination, 4 of the 5 claims are met. With every parameter at its set-point, the mean predicted XMuLV clearance of low-pH viral inactivation is 6.87 log10, which meets the claim of 6.6 log10. At the least favourable combination, with inactivation pH at its NOR high (3.6), hold time at its NOR low (60 min) and temperature at its NOR low (19 °C), it is 6.43 log10, which is below the claim. On

the model that INV-002 corrective action CA3 selects for XMuLV clearance, the claim is also not met at the least favourable combination of the NOR as characterized. For this reason, the low-pH viral inactivation report PCR-006 Version 2.0 moves the NOR and the set-point of inactivation pH lower and narrows the NORs of hold time and temperature. PCR-006 gives the values and the reasons.

Table 6.5: Viral clearance of each step against its platform claim (log10). Mean predicted LRF from the response surface model, with every parameter at its set-point and at the LFC.

Step	Virus	Claim	Mean LRF at set-points	Mean LRF at the LFC	Met
Low-pH viral inactivation	XMuLV	≥ 6.60	6.87	6.43	Not met
Anion exchange	XMuLV	≥ 5.50	6.58	5.83	Met
	MVM	≥ 4.00	4.87	4.20	Met
Virus filtration	MVM	≥ 4.62	5.55	5.16	Met
	XMuLV	≥ 4.62	6.10	6.04	Met

LFC, least favourable combination of NOR low, set-point and NOR high of the other parameters. LRF, log reduction factor.

6.4 Capability

In the re-analysis, the capability of acidic variants was assessed against both ends of its range (finding F4). The re-analysis uses the capability simulation that Version 1.0 reported, at the NOR as characterized. The lowest capability index of the drug substance moves from 1.51 for MVM clearance, as PCMR-001 Version 1.0 printed it, to 1.03 for acidic variants, against the lower limit of 18 %. PCMP-001 Version 2.0 sets a capability target of 1.33, which is common industry practice for a capable process. It sets a minimum of 1.00, the point at which a spread of three standard deviations reaches the criterion. Only the capability index of acidic variants falls below the target, and every attribute meets the minimum. The re-analysis also gives the probability of a drug substance batch outside specification for each attribute, from the same simulation. For acidic variants the simulation puts one batch in 2,000 outside the range, and a normal fit gives 971 per million. No simulated batch falls outside the criterion of any other attribute.

6.5 Fragments

In the re-analysis, fragments are judged on nrCE-SDS results only, against the fragment criteria at the Protein A, low-pH viral inactivation, cation exchange, anion exchange and virus filtration steps in Table 6.1. SEC fragment results from the Protein A pool and the low-pH viral inactivation pool will be reported for information only and

not used to judge a step. They could not have supported a fragment criterion at those steps, because they include host cell protein (finding F6).

6.6 Product impact

No GMP batch of A-Mab was manufactured or dispositioned on the basis of these documents, and Version 1.0 was not used in a regulatory submission. The findings concern how the characterization data were judged. The experiments and the drug-substance criteria that existed in Version 1.0 are unchanged. The process is unchanged except for NORs and set-points at the low-pH viral inactivation step, which PCR-006 Version 2.0 changes (§6.3). Version 2.0 adds one drug-substance criterion, for fragments, at ≤ 2.00 %. There is no product impact.

7 Corrective and preventive actions

The actions in Table 7.1 follow from the root cause and the contributing causes in §5. All 6 actions were open at closure and are tracked to their due dates in the CAPA system (International Council for Harmonisation 2008).

Table 7.1: Corrective (CA) and preventive (PA) actions. Every action was open at closure.

No.	Action	Owner function	Due
CA1	Reissue PCMP-001 with a priori intermediate acceptance criteria	MSAT	2026-09-30
CA2	Amend PCP-003 to PCP-010 with the criteria and the fragment method	MSAT	2026-09-30
CA3	Reissue PCR-003 to PCR-010, PTP-001, RA-001 and PCMR-001 at Version 2.0	MSAT	2026-09-30
PA1	Revise SOP-1002 so acceptance criteria are approved in the protocol before execution	Quality Assurance	2026-11-30
PA2	Add to the report review checklist of SOP-4001 a check that every acceptance criterion is traceable to the approved protocol	Quality Assurance	2026-11-30
PA3	Record execution dates and equipment calibration status in every study record	MSAT	2026-10-31

Corrective action CA1, the reissue of the master plan PCMP-001 with a priori criteria, removes the root cause for this campaign. It also answers the contributing cause that no document defined how much less a downstream step may clear within its NOR. PCMP-001 Version 2.0 fixes every step criterion before any study is judged against it, defines the downstream worst case each criterion protects against, and lets no step’s data set its own criterion. Corrective action CA2, the amendment of the step plans with the criteria and the fragment method, brings the decision AMD-001 to measure fragments by nrCE-SDS into the plans (finding F6). Corrective action CA3 reissues the reports. They judge each step against the new criteria, on the models selected under INV-002 (§2), and assess the capability of acidic variants against both ends of their range (finding F4).

Preventive action PA1, the revision of SOP-1002 so that acceptance criteria are approved in the protocol before execution, removes the root cause for future campaigns. It also removes the contributing cause that SOP-1002 did not require this. Preventive action PA2 adds to the report review checklist a check that every acceptance criterion can be traced to the approved protocol. That check would have detected the criteria added after approval (finding F1). Preventive action PA3 records execution

dates and calibration status in every study record. It answers the contributing cause that no second review checked the dates in the title blocks and study records (finding F5).

Two causes have no preventive action. No action adds a second review of whether a capability calculation uses the one-sided or two-sided criterion that the register gives. Corrective action CA3, which reissues the reports with a two-sided assessment of acidic variants, corrects finding F4 for this campaign only. For this product, the two-sided assessment is fixed in PCMP-001 Version 2.0, so the error cannot recur in this campaign. No action creates a procedure that would require a method decision to be assessed against the approved plans and put into effect by plan amendment either. Corrective action CA2 brings the decision AMD-001 into the plans, but a future method decision would again depend on the people involved to reach the documents.

8 Effectiveness check

The corrective actions are verified effective when the Version 2.0 package is effective and QA has repeated the review that triggered this investigation, on PCMR-001 Version 2.0, the master report. That review checks three things. Every step criterion used in a proven acceptable range analysis must appear in PCMP-001 Version 2.0 with its basis, and no criterion may be set by data from the step it judges. Every fragment result used to judge a step must be by nrCE-SDS. Every study must carry its execution window.

QA verifies the preventive actions by 2027-03-31. The check requires the next two characterization protocols approved with acceptance criteria and provenance, before execution. QA owns both checks.

9 Conclusion and closure

The investigation confirmed the classification as major. The criteria used to support the operating ranges in a BLA were not defined before the data were seen. The analyses that rested on them could therefore not be relied on as approved. The investigation found no product impact (§6.6).

The corrective actions are due with Version 2.0 on 2026-09-30, and the preventive actions by 2026-11-30. INV-001 was closed on 2026-09-18, with its remaining actions open and tracked to completion under the effectiveness check in §8.

References

CMC Biotech Working Group. 2009. “A-Mab: A Case Study in Bioprocess Development.” CASSS.

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